

WAUPACA, WI, USA

WMO: 720343

Lat: 44.333N

Lon: 89.020W

Elev: 252

StdP: 98.34

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54852

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-19.6	-27.4	0.3	-21.2	-25.0	0.4	-18.7	9.3	-6.1	8.3	-7.1	2.3	280	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	32.5	22.4	30.8	21.7	29.1	20.6	24.2	29.2	23.3	28.1	22.4	26.9	4.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.1	26.4	22.0	17.2	25.8	21.0	16.2	24.7	74.2	29.3	70.4	28.2	67.0	26.8	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.6	6.8	DB	-25.9	35.1	2.4	1.6	-27.6	36.2	-28.9	37.2	-30.3	38.1	-32.0	39.3	(4)
(5)			WB	-26.2	26.1	2.3	1.1	-27.8	27.0	-29.2	27.6	-30.5	28.3	-32.1	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-7.1	-6.0	0.8	7.4	14.4	19.7	22.1	21.0	16.6	9.5	2.4	-4.4	(6)	
	DBStd	11.51	6.48	6.20	6.46	4.97	4.98	3.49	3.25	3.06	4.43	4.80	5.24	5.97	(7)	
	HDD10.0	2147	531	447	295	107	14	0	0	0	4	69	232	447	(8)	
	HDD18.3	4069	789	680	546	329	143	25	6	9	83	276	477	706	(9)	
	CDD10.0	1454	0	0	8	29	150	291	374	340	202	55	5	0	(10)	
	CDD18.3	336	0	0	1	1	21	66	122	91	32	3	0	0	(11)	
	CDH23.3	3371	0	0	8	28	268	642	1248	825	317	35	0	0	(12)	
	CDH26.7	1076	0	0	1	5	82	191	459	247	88	3	0	0	(13)	
(14) Wind	WSAvg	2.7	2.9	2.9	2.9	3.3	2.9	2.4	2.1	1.9	2.1	2.7	2.8	2.8	(14)	
(15) Precipitation	PrecAvg	818	33	29	47	80	102	113	95	100	81	63	45	37	(15)	
	PrecMax	994	79	48	92	137	167	213	234	280	194	146	126	82	(16)	
	PrecMin	675	11	8	8	21	56	39	27	28	13	11	0	7	(17)	
	PrecStd	89	18	12	26	30	34	56	59	50	43	34	32	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	12.7	23.8	27.1	32.3	33.2	36.1	33.2	32.3	27.2	20.1	10.9	(19)	
		MCWB	4.5	8.4	16.4	15.3	20.1	22.2	24.0	22.5	22.8	19.8	13.4	6.7	(20)	
	2%	DB	4.9	7.0	17.3	22.5	28.8	30.9	32.6	31.1	29.4	23.7	15.6	6.7	(21)	
		MCWB	2.5	3.9	11.3	13.5	18.7	21.0	22.7	22.1	21.2	17.6	10.8	4.7	(22)	
	5%	DB	2.9	4.1	13.2	19.7	26.4	28.7	31.0	29.4	27.1	20.9	12.7	4.9	(23)	
		MCWB	1.6	1.8	8.6	11.7	17.2	20.1	22.0	21.0	19.5	15.8	9.0	3.4	(24)	
	10%	DB	1.4	2.2	9.9	16.7	23.8	27.2	29.2	27.7	24.8	17.8	10.1	2.8	(25)	
		MCWB	0.5	0.5	6.1	9.9	16.3	19.2	21.0	19.8	18.7	13.6	7.3	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	8.1	16.6	17.4	22.5	25.0	26.3	24.9	24.1	22.1	14.5	9.5	(27)	
		MCDB	7.7	12.7	23.4	23.0	28.8	30.3	32.4	29.6	29.0	25.5	18.0	10.5	(28)	
	2%	WB	3.1	4.2	12.7	15.1	20.7	23.1	24.5	23.8	22.7	19.3	11.7	5.2	(29)	
		MCDB	4.5	6.5	15.7	20.8	25.3	28.7	29.8	28.1	26.9	21.4	14.4	6.2	(30)	
	5%	WB	1.8	2.2	9.1	12.7	19.2	21.9	23.4	22.7	21.5	16.6	9.5	3.4	(31)	
		MCDB	2.8	4.1	13.4	17.9	24.4	26.5	28.4	26.9	25.0	20.0	12.5	4.6	(32)	
	10%	WB	0.6	0.5	6.1	10.3	17.5	20.7	22.5	21.8	20.0	14.1	7.5	1.7	(33)	
		MCDB	1.4	1.8	9.7	15.8	22.1	24.6	27.2	25.8	23.2	17.3	9.5	2.7	(34)	
(35) Mean Daily Temperature Range	MDBR		8.4	9.2	10.3	11.3	12.4	11.8	12.8	11.9	12.4	10.5	8.6	7.4	(35)	
	5% DB	MCDBR	9.1	10.3	14.5	17.4	15.6	14.0	14.3	13.4	14.7	14.8	13.5	8.6	(36)	
		MCWBR	7.5	7.8	8.8	9.2	7.1	6.2	5.9	5.4	7.1	8.9	9.0	7.2	(37)	
	5% WB	MCDBR	8.4	9.6	13.7	15.5	13.1	11.9	12.4	11.1	11.9	12.6	12.3	7.8	(38)	
		MCWBR	7.2	7.4	8.9	9.3	6.8	6.4	6.1	5.4	6.5	9.0	8.9	6.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.258	0.309	0.327	0.364	0.399	0.403	0.392	0.393	0.364	0.324	0.293	0.266	(40)	
	taud		2.400	2.237	2.366	2.330	2.275	2.298	2.337	2.327	2.403	2.511	2.518	2.443	(41)	
	Ebn at Noon		880	872	910	898	872	868	873	859	858	856	833	833	(42)	
	Edh at Noon		73	104	106	120	131	128	122	119	102	80	66	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.66	2.75	3.89	4.54	5.45	5.96	6.11	5.22	4.16	2.63	1.68	1.23	(44)	
	RadStd		0.19	0.25	0.42	0.46	0.42	0.43	0.30	0.34	0.32	0.28	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page